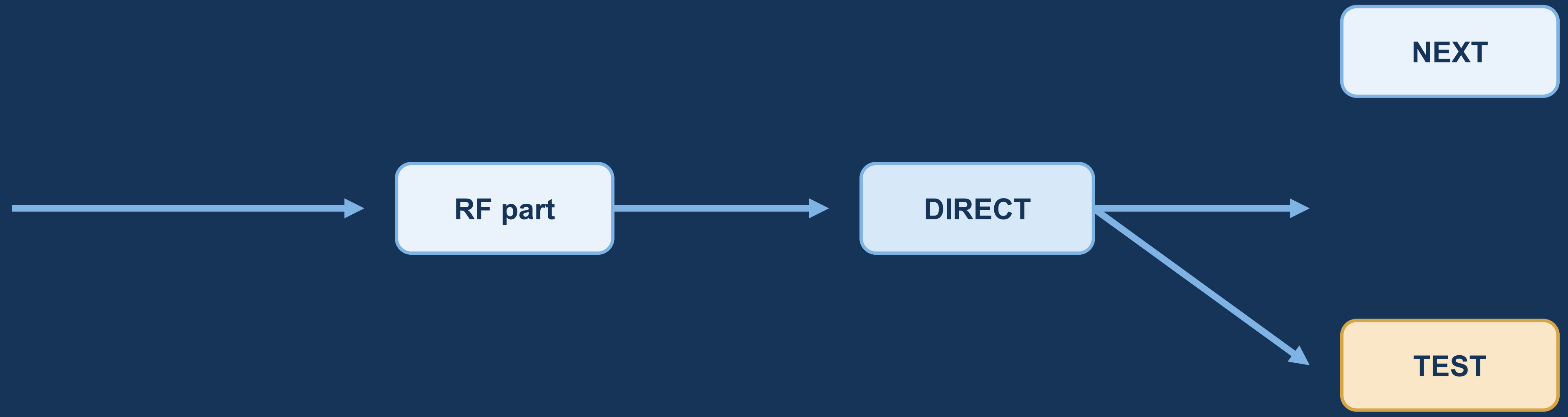
ENGINEERING DESIGN REVIEW

12 GHz RF jumpers

How to change the path from direct-through to a test connector

Architecture options, RF tradeoffs, layout rules, and a verification plan

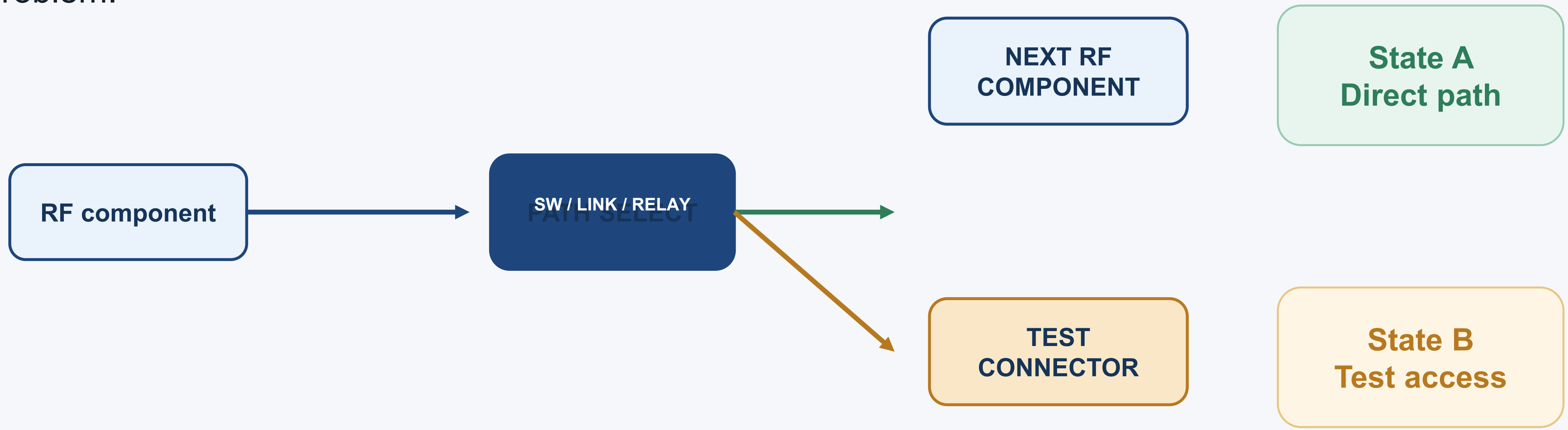


Prepared for technical discussion

KEY QUESTION

The design question is a controlled RF state change

One node must support two mutually exclusive destinations without turning the unused branch into an RF problem.



- Is the test port a replacement path or a monitor port?
- Must switching occur powered, unpowered, or only during service?
- What are the allowed loss, isolation, VSWR, and lifetime?

DESIGN TARGETS

At 12 GHz, the interconnect is part of the circuit

25 mm	~15-18 mm	1 mm	-20 dB
free-space wavelength	typical guided wavelength	is already a meaningful discontinuity	a useful first isolation target

What creates risk

A jumper or test feature adds length, pads, vias, connector launch geometry, and an unused branch. Each can create reflection and loss.

What must be measured

- S-parameters across the full operating band
- Insertion loss in both selected states
- Return loss at component, switch, and test ports
- Isolation from the selected path to the unused path

Do not choose the topology from DC continuity alone. Choose it from the RF error budget and the operating procedure.

There are two fundamentally different requirements

What does "test connector" need to do?

Replace the next component for a measurement

Observe the signal while the next component stays connected

Use a 1:2 selector

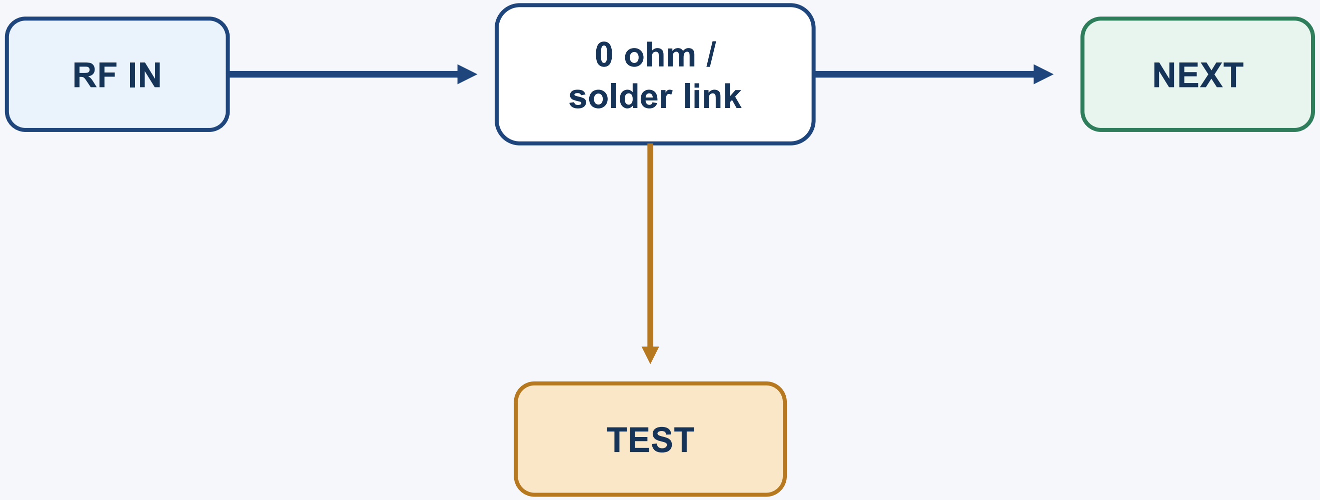
- Direct path is open when TEST is selected
- Isolation protects the inactive destination
- Best when test means "break in here"

Use a coupler or splitter

- Main path remains active
- Test port receives a coupled, lower-power sample
- Best for non-invasive monitoring or calibration

LOW COST

Option 1: fixed PCB links are cheap, but not operationally flexible



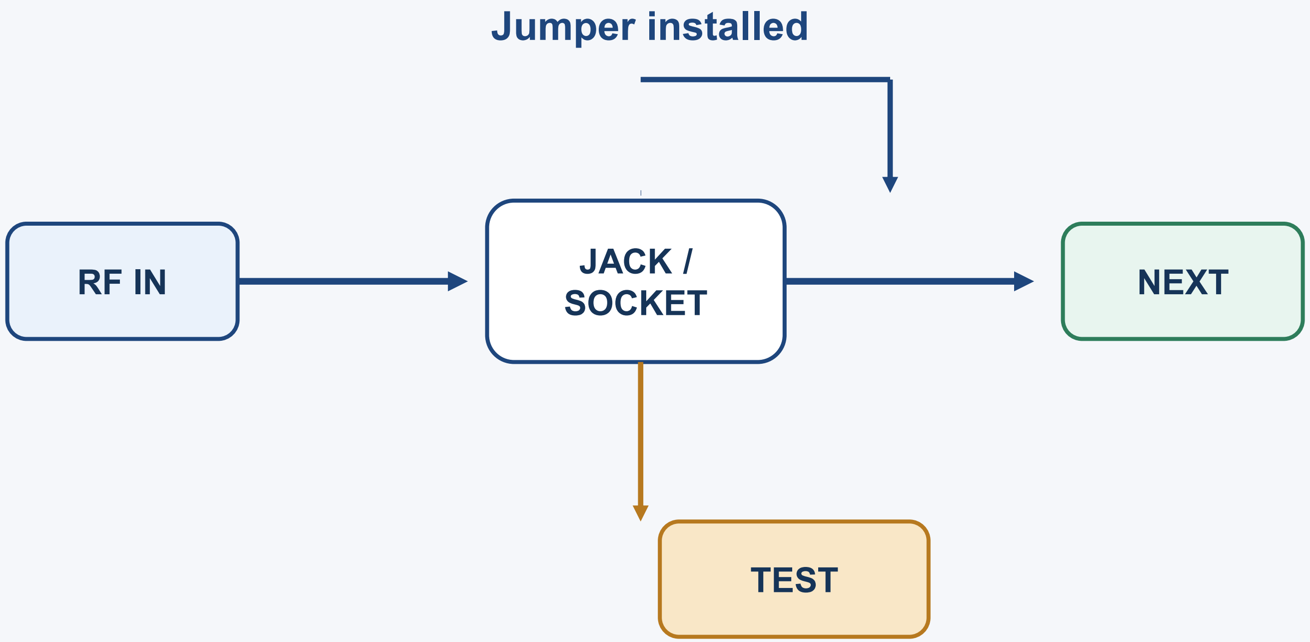
Assembly variant

- Solder bridge or 0 ohm resistor selects one route
- Coax pigtail or edge connector can expose the test node
- No live switching; service requires rework or a build variant
- Unused branch must be terminated, removed, or kept very short

Use when: low unit volume, factory configuration, and rework is acceptable. Avoid when technicians must switch repeatedly.

SERVICEABLE

Option 2: a mechanical RF jumper gives service access without active electronics

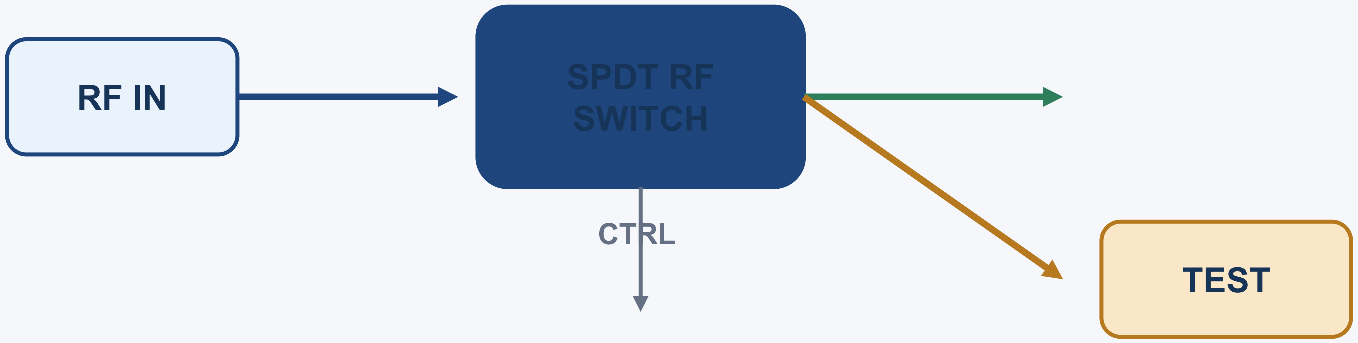


- Options: 2-port coax jumper, removable shunt, or board-to-board RF cable
- Excellent passive linearity and power handling when well matched
- Connector launch and repeated mating dominate performance
- Need keyed access, ESD handling, strain relief, and a defined torque/process

Best fit: occasional field or lab switching where the RF path must be physically obvious.

RECOMMENDED DEFAULT

Option 3: an RF switch is the cleanest repeatable 1:2 selector



- Series SPDT: low bill of materials, finite isolation, control sequencing required
- Reflective vs absorptive: decide how the unused throw is terminated
- Choose switch family for 12 GHz, not just “microwave capable”
- Check insertion loss, isolation, P1dB / IP3, ESD, control voltage, and switching lifetime

Default architecture for a production product: place the switch at the branch point and keep both routes 50 ohm.

WHEN TO USE

Option 4: relays, MEMS, and external instruments trade speed for RF quality

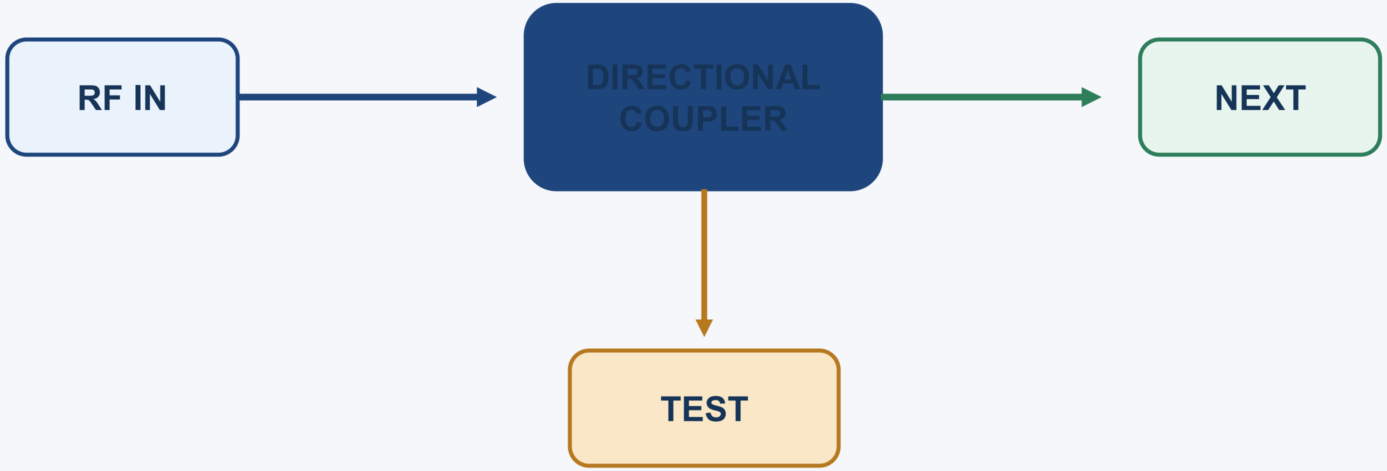
Architecture	RF strength	Main penalty	Good fit
Electromechanical relay	Very good isolation / linearity	Slow; wear; coil drive; size	Power-off state, high isolation, low switching count
RF MEMS switch	Low loss; high linearity; low power	Specialized; control/packaging; slower than IC	High-performance instrumentation or niche products
Coaxial RF switch box	Excellent connectorized RF; easy debug	Cost, volume, external wiring	Lab fixture, characterization, prototype
Probe / pogo / VNA fixture	No permanent connector on product	Repeatability, contact force, calibration	Manufacturing test, not normal operation

A relay can be the right answer when the RF budget is tight and switching is infrequent. An IC switch is usually the better product compromise.

ALTERNATIVE

If the test connector must monitor, do not switch the main path

Coupled sample



- Main signal remains connected to NEXT in every state
- Coupling factor sets test-port power and measurement dynamic range
- Add a switch or attenuator at TEST if multiple instruments share the port
- Terminate unused test paths with a good 50 ohm load

Use this when "test" means observe, not replace. It avoids interrupting the system path, but adds permanent insertion loss and a coupler footprint.

The RF layout determines whether the chosen topology survives 12 GHz

1	**Launch cleanly**	Controlled-impedance geometry; short pad transitions; connector footprint from the vendor.
2	**Keep the branch compact**	Place switch / link at the decision node. Minimize the open stub to the inactive route.
3	**Protect the return path**	Continuous reference plane; via fence around the RF line; avoid unnecessary layer changes.
4	**Plan for calibration**	Put test connector and calibration reference plane where the VNA can actually de-embed to.
5	**Control mechanics**	Connector torque, mating cycles, ESD, cable bend radius, and access direction are RF requirements.

At 12 GHz, “short” is an RF statement: compare physical length to guided wavelength and simulate or measure the launch.

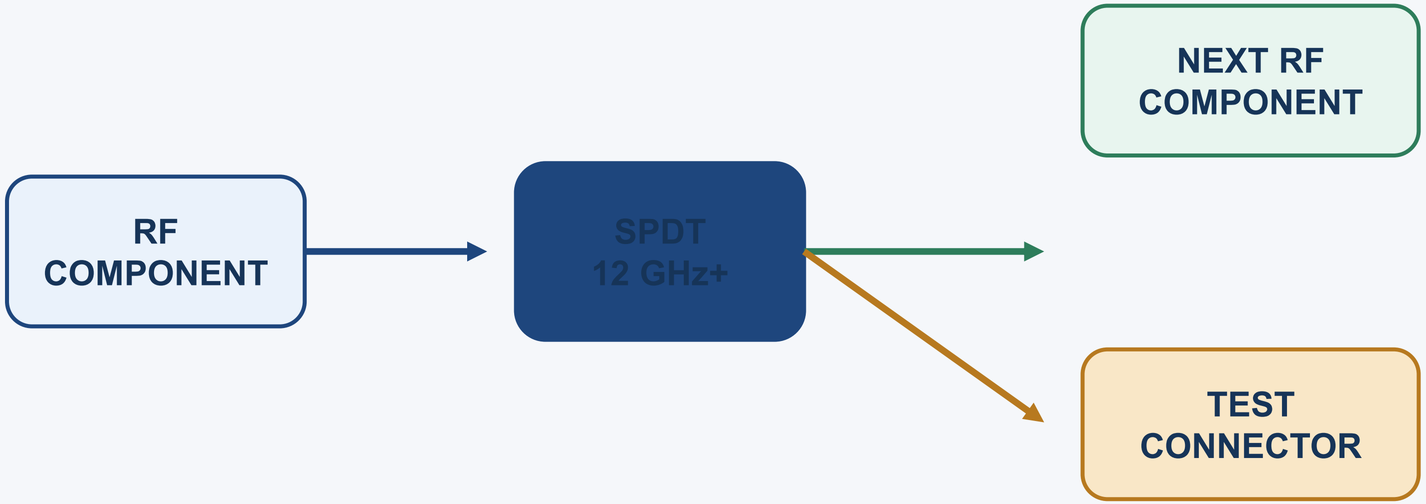
DECISION GUIDE

Select the topology by the dominant constraint, not by habit

Need	Preferred architecture	Why	Watch-outs
Factory variant	Solder link / 0 ohm	Lowest cost and loss if route is fixed	Rework and unused branch
Occasional service	Mechanical coax jumper	Visible, passive, easy to understand	Mating and connector launch
Repeatable product switching	SPDT RF switch	Fast, controlled, compact	Loss, isolation, control state
Maximum isolation	EM relay / coax switch	Strong off-state behavior	Size, speed, wear, cost
Continuous monitoring	Directional coupler	No interruption of main path	Permanent loss and sample level
Manufacturing test	Pogo / probe fixture	No product connector required	Contact repeatability and calibration

BASELINE DESIGN

A practical baseline is an SPDT switch plus a connectorized test route



Default state: DIRECT
Test state: TEST

- Use a switch specified beyond 12 GHz with characterized S-parameters
- Keep both switch throws 50 ohm and route the test throw to a robust connector
- Define break-before-make or make-before-break behavior
- Add interlock / state indication so the tester knows which path is active

If the requirement is observation rather than substitution, replace the SPDT with a directional coupler and keep NEXT permanently connected.

EXIT CRITERIA

Verification must prove both RF performance and real-world usability

01 → **02** → **03** → **04** → **05**

01 Simulate

EM model of switch / link, pads, vias, connector launch, and inactive branch.

02 Calibrate

VNA calibration at the intended reference plane; document adapters and de-embedding.

03 Measure

S11 / S22, insertion loss, isolation, phase, and power handling across the full band.

04 Stress

Mating cycles, torque, cable movement, temperature, vibration, ESD, and state transitions.

05 Release

Freeze footprint, control truth table, connector process, acceptance limits, and test record.

Minimum decision package

Operating band | max loss | return loss | isolation | power | switching mode | mating life | environmental profile

Recommendation: start with the SPDT baseline, then downgrade to a passive link only if the use case is truly configuration-only.

NEXT STEP

Takeaway: define "test" first, then choose the jumper

Replacement path →	SPDT / relay / mechanical jumper
Monitor path →	Directional coupler / splitter
Factory-only variant →	Solder link / 0 ohm / PCB option
Manufacturing access →	Probe or pogo fixture

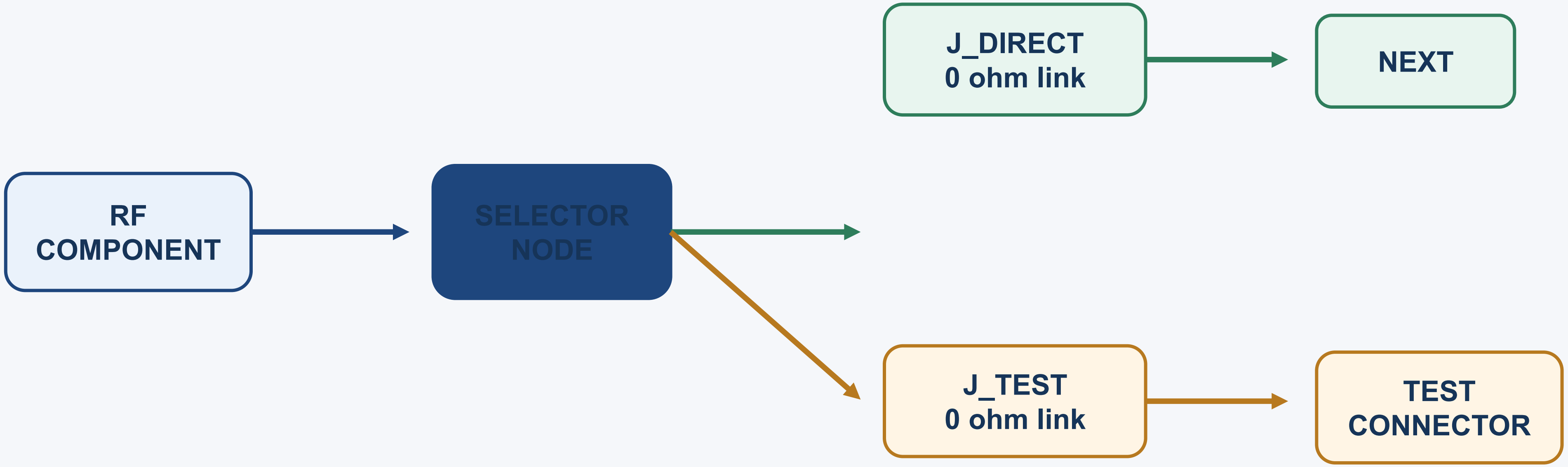
The robust 12 GHz design is the one that makes the inactive path electrically quiet, mechanically repeatable, and measurable.

Open design inputs: frequency span, power, connector family, switching frequency, environment, and whether "test" replaces or monitors the next component.

LOW-COST PATH

For a bachelor thesis, treat the 1:2 selector as an assembly option

The selector is not changed while powered. You change which RF link is populated, then measure the new configuration.



**Populate exactly one link at a time:
J_DIRECT OR J_TEST, never both.**

This is a reconfigurable prototype, not a live RF switch.

Use three build states: direct, test, and safe open

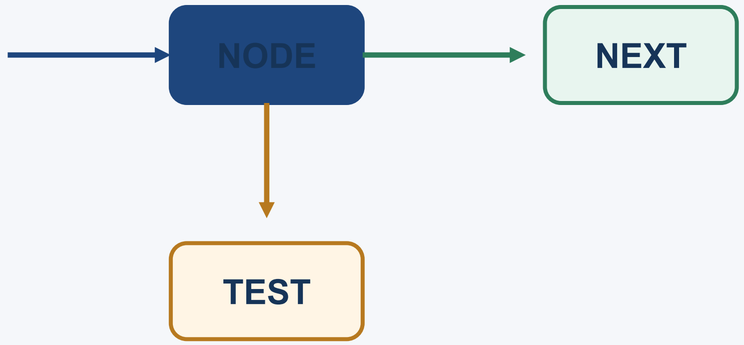
Build state	J_DIRECT	J_TEST	Connector	Purpose
A: Direct / final	POPULATED	DNP	DNP or 50 ohm	Normal signal path to the next component.
B: Test / measure	DNP	POPULATED	POPULATED	Route the component output to the VNA or test instrument.
C: Safe open	DNP	DNP	DNP	Debug / isolation state; no complete path.

- DNP = do not populate; it is an assembly instruction, not an electrical component.
- Keep one spare PCB or a small rework area because repeated soldering can damage pads.
- Record the exact population state on every measurement file and PCB photograph.

DO NOT LEAVE A STUB

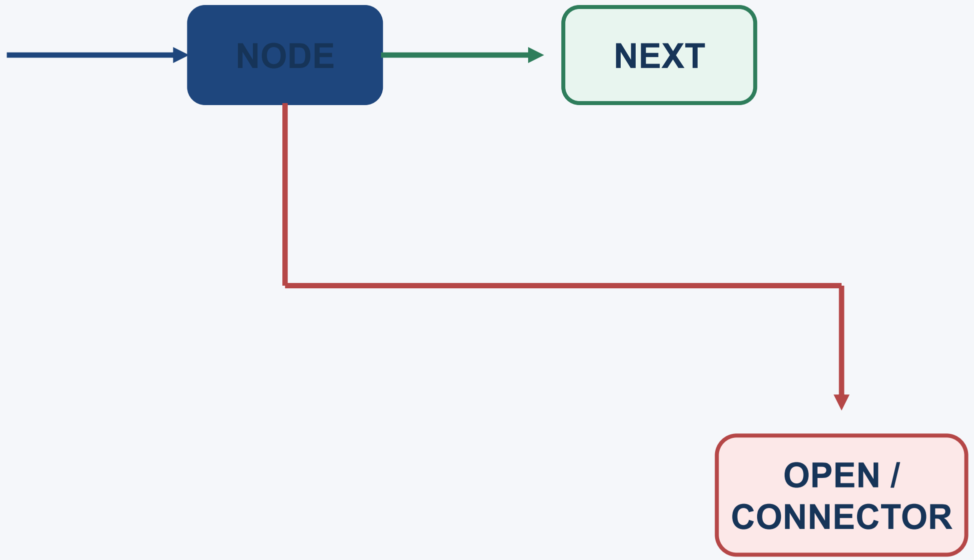
The low-cost selector works only if the inactive branch is controlled

Good: branch point is compact



Keep the distance from NODE to either link as short as possible.

Bad: long open branch



An unselected 50 ohm branch is not automatically harmless. Terminate it, remove it, or make the stub electrically very short.

PRACTICAL BUILD

Use a connector only during characterization, then depopulate or terminate it

1 Characterization PCB → 2 Measure components → 3 Commit final path

1. Characterization PCB	2. Measure components	3. Commit final path
J_DIRECT DNP J_TEST POPULATED Test connector fitted	VNA -> TEST Measure each block Save S-parameters	J_DIRECT POPULATED J_TEST DNP Connector DNP or 50 ohm

- Do not treat the test connector as “free” once it is left open: the connector launch and branch still load the circuit.
- If the connector remains fitted in the final board, connect a suitable 50 ohm termination when the port is unused.
- If budget allows only one board spin, put the test connector on a small detachable coupon or a breakaway edge section.

VNA METHOD

Measure one block at a time, but always define the reference plane

01	**Calibrate**	SOLT or TRL at the connector / fixture reference plane
02	**Connect**	Use short, known-good 50 ohm cables and torque connectors consistently
03	**Measure**	S11, S21, and phase across the complete frequency span
04	**Compare**	Check against the same fixture and cable baseline
05	**Record**	File name includes PCB ID, build state, component ID, and date

The measurement is only useful if the fixture contribution is known. First measure a through / thru standard or a known-good reference path.

DATA RECORD

A simple test matrix makes the thesis results traceable

Item	Build state	What to measure	Expected question
UUT-1	B: TEST	S11 / S21	Does the component meet its datasheet behavior at 12 GHz?
J_TEST path	B: TEST	S11 / S21 / phase	How much does the test link and connector add?
J_DIRECT path	A: DIRECT	S11 / S21	Does the final route preserve the direct path?
Inactive branch	A: DIRECT	S11 / S21 at NEXT	Does the unused test branch disturb the direct route?
Repeatability	A and B x 3	S21 spread	Does rework or cable movement change the result?

Use the same frequency grid, IF bandwidth, power level, calibration, and cable routing for comparable results.

Choose the 0 ohm implementation as an RF part, not a generic resistor

PART SELECTION

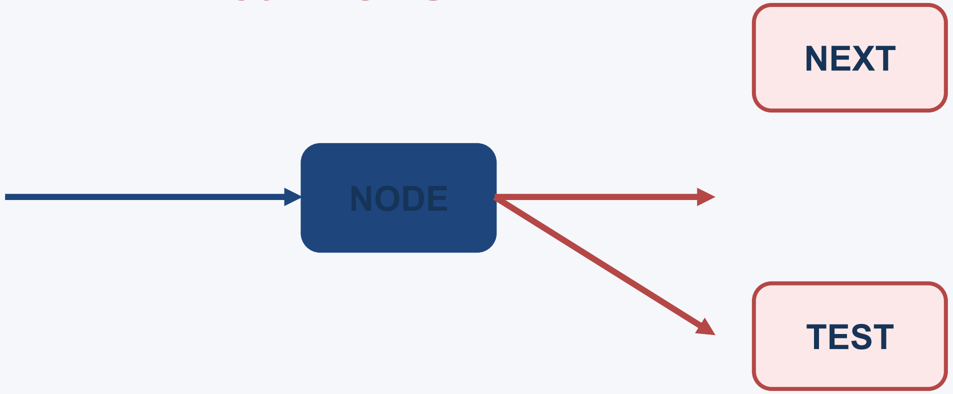
Package	Prefer a small package with a short current path; 0402 is easier to assemble, 0201 can reduce parasitic size.
Footprint	Use a symmetric 50 ohm launch into the link; avoid long rectangular pads that widen the line.
Solder bridge	Lowest BOM cost, but manual solder volume and repeatability are harder to control.
Rework	Add access, courtyard, and nearby ground; plan for wick / hot-air rework without lifting pads.

Important: a 0 ohm resistor is not an ideal RF short. Verify the chosen part and footprint with S-parameter data or measure the assembled link.

CHECK BEFORE POWER

Do not populate both links: that creates a splitter, not a selector

Both POPULATED



- The RF component now sees two destinations in parallel.
- Power splits and the two loads interact through the branch.
- A test connector cable can change the final path response.
- Document a visual inspection or continuity check before every VNA run.

Assembly rule: exactly one of J_DIRECT and J_TEST is populated for a valid 1:2 build state.

FINAL PROPOSAL

Thesis-ready recommendation: one low-cost board, two controlled builds

Board design → **Characterize** → **Finalize** → **Evidence**

Board design

- 50 ohm RF traces
- Compact branch node
- J_DIRECT and J_TEST footprints
- Test connector footprint with DNP option
- Clear silkscreen: DIRECT / TEST

Characterize

- Populate J_TEST
- Measure each component
- Measure test route
- Save raw Touchstone data
- Compare against datasheet

Finalize

- Populate J_DIRECT
- DNP J_TEST
- DNP or terminate connector
- Repeat final measurement
- Report difference

Evidence

- Photos of both states
- BOM population table
- VNA calibration record
- S11 / S21 plots
- Uncertainty and limitations

This gives you a defensible thesis story: measure the device, quantify the test path, then quantify the impact of the final direct path.

MODEL HIERARCHY

Design the selector in four models, from simple to realistic

1. **Ideal circuit** → Check the logic: one input, two mutually exclusive outputs.
2. **Transmission-line model** → Check 50 ohm lines, branch length, pads, and connector launches.
3. **EM model** → Capture coupling, discontinuities, vias, and the real ground plane.
4. **Measured board** → Compare S-parameters and update the model assumptions.

Do not start in KiCad by drawing a beautiful board. Start by proving the RF topology and dimensions in AWR, then transfer the controlled geometry deliberately.

INPUTS

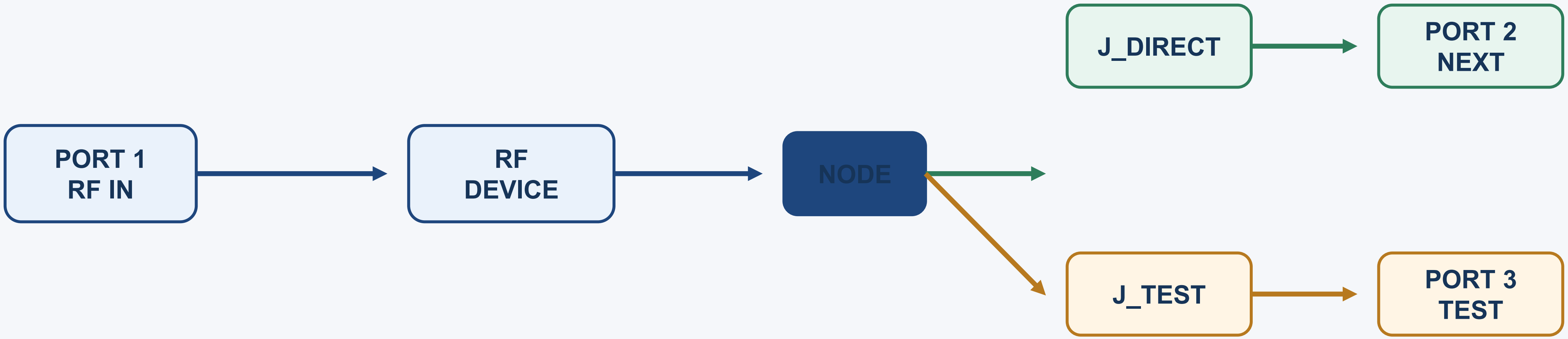
Step 1: freeze the design inputs before opening the simulator

- **01 Frequency** — Start / stop frequency, for example 10 to 14 GHz if 12 GHz is the center.
- **02 Impedance** — 50 ohm system; define acceptable S11, S21, and isolation limits.
- **03 Stack-up** — Substrate Dk, loss tangent, thickness, copper thickness, soldermask assumption.
- **04 Components** — RF component package, pad dimensions, 0 ohm link package, connector footprint.
- **05 States** — State A: J_DIRECT populated. State B: J_TEST populated. Never model both as the valid state.
- **06 Boundary** — Define the VNA reference plane: board connector, component pin, or de-embedded plane.

Ask the PCB manufacturer for the controlled-impedance stack-up before selecting trace widths. Do not use a generic calculator value as the final dimension.

FIRST SIMULATION

Step 2: build the ideal 1:2 selector schematic in AWR



- Use ideal ports and an ideal switch/link first to verify the signal-flow logic.
- Simulate State A and State B as separate schematics or separate link values.
- Plot S21 for the selected route and S31 for the alternate route; confirm the inactive route is not accidentally connected.

MSUB + MLIN

Step 3: replace ideal wires with AWR microstrip models

- **MSUB**: Define substrate and copper stack-up
- **MLIN**: Set width, length, metal thickness, and substrate
- **MTEE**: Model the branch junction
- **MSTEP / MOPEN**: Model width transitions and open ends where applicable
- **PORT**: Set reference impedance and calibration reference

Parameterize before optimizing: line width W, branch length Lbranch, pad length Lpad, pad width Wpad, and connector transition dimensions.

S-PARAMETERS

Step 4: add realistic component and connector data

RF component	Use vendor S2P file if available. Check port order, impedance, frequency span, and de-embedding plane.
0 ohm link	Start with an ideal short, then replace it with vendor S-parameters or a measured coupon.
RF connector	Use a vendor model, launch model, or a measured connectorized reference.
Cables / adapters	Do not hide them in the DUT model. Calibrate or include their S-parameters explicitly.

- Touchstone files normally use a .s1p, .s2p, or similar extension.
- Confirm whether the file is measured at 50 ohm and whether it uses dB/angle or real/imaginary format.
- Keep a copy of every model used in the thesis project folder and record its source.

OPTIMIZE

Step 5: simulate both valid population states and sweep the sensitive dimensions

Case	Link definition	Plots	Decision
State A	J_DIRECT = short; J_TEST = open / terminated	S11 at input; S21 to NEXT; effect of inactive branch	
State B	J_DIRECT = open; J_TEST = short	S11 at input; S21 to TEST; test-route loss	
Sensitivity	Sweep W, Lbranch, Lpad, and connector launch	Find which dimension moves the 12 GHz response most	
Acceptance	Add limits to the graphs	Pass / fail by frequency, not only at one marker	

Use optimization only after the model is physically sensible. An optimizer can find a mathematically good but unmanufacturable geometry.

AXIEM-STYLE EM

Step 6: use EM simulation for the branch, pads, and return path

01	**Create EM structure**	Select the selector region, stack-up, ports, and ground boundary.
02	**Draw physical metal**	Include trace widths, branch, pads, vias, connector launch, and nearby copper.
03	**Set ports**	Place ports where the circuit model will connect; keep the reference plane intentional.
04	**Sweep frequency**	Use the same frequency range as the schematic simulation.
05	**Export S-parameters**	Use the EM result as a block in the circuit schematic.

EM is where the model begins to see the real branch discontinuity. Start with a small region around the selector so the simulation remains fast and interpretable.

CLOSE THE MODEL

Step 7: co-simulate the EM selector with the RF component

- Replace the ideal branch in the circuit schematic with the EM-generated multiport data.
- Run State A and State B with the same RF component and port definitions.
- Compare circuit-only, EM-only, and co-simulated results to isolate where the loss or mismatch comes from.
- Save the exact EM project, mesh settings, stack-up, and exported file used for the thesis plot.

If the measured board disagrees, first check stack-up and reference planes before changing the topology.

PCB IMPLEMENTATION

Step 8: transfer the validated dimensions into KiCad deliberately

1	**Create the project**	Set units, grid, net classes, board outline, and revision-controlled project folder.
2	**Enter the stack-up**	Use the manufacturer stack-up, dielectric thickness, Dk, copper thickness, and impedance rules.
3	**Place footprints**	Place RF component, link footprints, test connector, and nearby ground vias first.
4	**Route RF**	Use the AWR width and length as starting values; keep branch short and reference plane continuous.
5	**Add fabrication details**	Silkscreen population labels, DNP notes, polarity, connector direction, and test points.
6	**Run checks**	DRC, courtyard, annular ring, soldermask, fab drawing, and impedance review.

Do not assume AWR and KiCad use identical origin, layer naming, or pad geometry. Compare the actual copper dimensions after layout.

PRE-FAB

Step 9: perform a layout review before fabrication

Geometry	Are W, Lbranch, pad dimensions, and connector launch close to the AWR model?
Ground	Is the reference plane continuous under the RF path? Are ground vias close enough to the transition?
Assembly	Can J_DIRECT and J_TEST be reworked without damaging the RF trace? Are DNP states unambiguous?
Mechanical	Can the test cable mate without bending the board or touching sensitive parts?
Documentation	Are stack-up, footprints, link values, DNP notes, and revision recorded?

Export a PDF of the final layout and annotate the RF current path. This is useful in the thesis and catches accidental detours.

MODEL VS HARDWARE

Step 10: validate the fabricated board and update the model

AWR prediction →	KiCad release →	Board measurement →	Correlation →	Thesis result
S11 / S21 / isolation plots with model version	Gerbers, drill files, stack-up, BOM, population state	Calibration, photographs, exact cable routing, raw Touchstone files	Overlay measured and simulated traces; explain differences	Final path choice, error budget, limitations, and next revision

The goal is not a perfect first simulation. The goal is a traceable loop from assumptions to geometry to measurement.

PROJECT PLAN

A manageable thesis schedule is: model, simulate, layout, measure, explain

Week 1	**Requirements + stack-up**	Confirm frequency span, manufacturer capability, connector, RF component, and measurement equipment.
Week 2	**AWR schematic**	Build ideal and transmission-line models; define State A and State B.
Week 3	**AWR EM**	Model branch, pads, vias, connector launch; run sensitivity sweep.
Week 4	**KiCad layout**	Implement the selected dimensions, add DNP notes, review and release fab files.
Week 5	**Board measurement**	Calibrate VNA, characterize State B, then rework to State A and repeat.
Week 6	**Correlation + thesis**	Overlay plots, explain mismatch, document rework and final recommendation.

Recommended first action: obtain the PCB stack-up and the RF component S-parameter file, then build the ideal State A / State B schematic in AWR.